

Cluster Robust Variance Estimation

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1 Introduction

In constructing the confidence intervals of the OLS estimator $\hat{\beta}_{\text{ols}}$, it is the asymptotic distribution of

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i$$

that determines the estimator's asymptotic distribution. Assuming that $E[X_i u_i] = 0$, the Lindernberg-Levy CLT apply:

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{N} \sum_i \text{Var}(X_i u_i)\right)$$

But this hold true only if the individual observations are all independent, in fact summing 2 random variables X_1 and X_2 we get another random variable (call it Z) with expected value equal to $E[X_1] + E[X_2]$ and variance equal to $\text{Var}(X_1) + \text{Var}(X_2) + 2 \cdot \text{cov}(X_1, X_2)$. Summing N random variables $(X_1, \dots, X_n)$, assume $E[X_i] = 0$, we get a new random variable $Z_N = \sum_{i=1}^N X_i$:

$$Z_N \sim \left(0, \sum_{i=1}^N \text{Var}(X_i) + 2 \sum_{i \neq j}^N \sum_j^N \text{cov}(X_i, X_j)\right)$$

if all the X_i s are **independent** all the joint moments (and so also the covariancers) are equal to zero and, dividing by $\sqrt{N}$:

$$\frac{1}{\sqrt{N}}(Z_N) \sim \left(0, \frac{1}{N} \sum_{i=1}^N \text{Var}(X_i)\right)$$

So the variance of Z_N scaled by $\frac{1}{\sqrt{N}}$ is the average variance of all the X_i s. But here comes the problem, if our data are clustered we do not have that all the observation are independent, but rather we have that the observation of all the individuals inside a cluster are correlated, so we would have that

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i u_i \xrightarrow{d} \mathcal{N} \left(0, \frac{1}{N} \sum_{i=1}^N \text{Var} (X_i u_i) + \frac{2}{N} \sum_{i \neq j}^N \sum_{j=1}^N \text{cov} (X_i u_i, X_j u_j) \right)$$

The variance estimator that is most commonly used, the White variance estimator assumes that all the covariances in the expression above are 0, and so estimates only $\frac{1}{N} \sum_{i=1}^N \text{Var} (X_i u_i)$, so it would not consistently estimate the asymptotic variance.

1.1 Why we estimate $\text{Var} (\hat{\beta})$

To follow what comes next, it is essential to clarify **what we are actually estimating** when we estimate the variance of

$\hat{\beta}_{\text{ols}}$. After all, $\hat{\beta}_{\text{ols}}$ is just a number—so what does it mean to speak of its variance?

Is it a random variable? Does it have a distribution? Yes. The distribution we refer to is the distribution of $\hat{\beta}_{\text{ols}}$ **across repeated samples**, i.e., the values it would take if we changed the individuals drawn from the population. Since $\hat{\beta}_{\text{ols}}$ is a function of random variables (X_i and u_i), if we knew their full population distributions, we would know the exact value of β . But because we only observe a sample, our estimate varies from sample to sample. In this sense, $\hat{\beta}_{\text{ols}}$ is a random variable and therefore has a distribution.

1.2 What are clustered data?

Let's take a step back and ask ourselves what does it mean that our data are clustered? A. C. Cameron and D. L. Miller [1] define a cluster as a subgroup in the population such that individuals inside it has errors that are correlated. According to A. Abadie, S. Athey, G. W. Imbens, and J. M. Wooldridge [2] what matters is the correlation of $X_i u_i$. This difference arises due to the fact that in A. C. Cameron and D. L. Miller [1] regressors are considered non stochastic. In essence, the defining feature of a cluster is that individuals within it exhibit some form of **unobserved** correlation.

Suppose our sample contains N individuals, each with an observed trait X . We can represent these as $X_{ig} = (X_{11}, \dots, X_{N_1 1}, X_{12}, \dots, X_{N_2 2}, \dots, X_{N_G G})$, where i indexes individuals within a cluster and g indexes the cluster.

We can think of all these X_{ig} as being drawn from the same population distribution, so they are **identically distributed**. Let us assume this distribution has mean 0.

However, individuals belonging to the same cluster are **not independent**. Their joint distribution is not simply the product of their marginals, their covariances may not be zero.

Saying that observations within a cluster are correlated means that knowing the value of one individual gives information about another individual in the same cluster. If individuals i and j belong to the same cluster g , then:

$$P(X_i > E[X] \mid X_j > E[X], i, j \in g) > P(X_i > E[X])$$

and similarly:

$$P(X_i < E[X] \mid X_j < E[X], i, j \in g) > P(X_i < E[X]).$$

outcomes inside the same cluster tend to move together relative to the overall mean—they are positively associated. If we focus on a single cluster g , we can view its observations as:

$$X_g = \{X_i \mid \text{individual } i \in \text{cluster } g\}$$

Each cluster is a subpopulation with its own distribution, and the overall population is the mixture of all G cluster distributions.

1.3 General setup

In this section, we lay out the general setup that will guide the rest of the thesis. We assume that there are heterogeneous effect across clusters, this means that the true parameter β is different in every cluster.

$$y_{ig} = \alpha + \beta_g X_{ig} + u_{ig}$$

We model u_{ig} as having a cluster component and an individual-and-cluster component:

$$u_{ig} = c_g + v_{ig}$$

Where $v_{ig} \sim (0, \sigma_v^2)$, $c_g \sim (0, \sigma_c^2)$ but of course c_g varies only at the cluster level. In this way the expected value of the error given that individual i is in cluster g is:

$$\mathbb{E}_g(u_{ig}) = c_g$$

But as long as we have an intercept we can assume w.l.o.g that $\mathbb{E}[u_{ig}] = 0$. If I want to estimate β_g in each cluster g , I get (in each cluster):

$$\hat{\beta}_g = \beta_g + \left(\sum_{i \in g} X_i^2 \right)^{-1} \left(\sum_{i \in g} X_i u_i \right)$$

And:

$$\hat{\beta}_g \stackrel{a}{\sim} \mathcal{N} \left(\beta_g, \frac{1}{N_g} \left(\sum_{i \in g} X_i^2 \right)^{-1} B \left(\sum_{i \in g} X_i^2 \right)^{-1} \right)$$

where B_g is:

$$B_g = \text{Var} (X_{ig} u_{ig})$$

If we estimate β separately across G clusters, we obtain a vector of cluster-specific estimates $\{\hat{\beta}_g\}_{g=1}^G$.

If for each cluster we could consistently estimate its asymptotic variance, we could take the average of these cluster-level estimates to obtain a consistent estimate of the asymptotic variance

of the overall $\hat{\beta}_{\text{ols}}$, since a linear combination of normal random variables is also normal. But first we have to ask ourselves whether each of the estimated parameters is consistent.

1.4 What is a cluster?

But let's take a step back and ask ourselves what does it mean that our data are clustered?

A. C. Cameron and D. L. Miller [1] define a cluster as a subgroup in the population such that individuals inside it have errors that are correlated. According to A. Abadie, S. Athey, G. W. Imbens, and J. M. Wooldridge [2] what matters is the correlation of $X_i u_i$. This difference arises only due to the fact that in A. C. Cameron and D. L. Miller [1] regressors are considered non stochastic.

In essence, the defining feature of a cluster is that individuals within it exhibit some form of unobserved correlation.

Qualche esempio

Qua devo aggiungere gli esempi dalla letteratura tipo:

- B. R. Moulton [3]
- M. Bertrand, E. Duflo, and S. Mullainathan [4]

Let's say that in our population there are N individuals from which we observe a trait X , so that we have N random variables $X_{ig} = (X_{11}, \dots, X_{Ng1}, X_{12}, \dots, X_{Ng2}, \dots, X_{NgG})$, where i indexes an individual inside a cluster and g is the cluster. We can think of them as being drawn from the same distribution (so that they would be identically distributed), let's assume that this distribution has mean 0.

But they are not independent if they share the same cluster. This means that their joint distribution is not just the product of the marginals and their covariance may not be 0, so the convolution of two of them in the same cluster is:

$$X_{11} + X_{12} \sim (0, \text{Var}(X_1) + \text{Var}(X_2) + 2 \text{Cov}(X_1, X_2))$$

Saying that observations within a cluster are correlated means that the values of one individual carries information about the values of another one in the same cluster. More concretely, if two individuals i and j belong to the same cluster g , then observing that one of them takes a value above the overall (population-wide) mean of X increases the likelihood that the other will also be above the mean:

$$P(X_i > E[X] \mid X_j > E[X] \text{ and } i, j \in g) > P(X_i > E[X]).$$

In other words, the event that X_j exceeds the population expectation $E[X]$ raises the conditional probability that X_i does the same when both belong to the same cluster. This would be the same for values below the mean:

$$P(X_i < E[X] \mid X_j < E[X] \text{ and } i, j \in g) > P(X_i < E[X]).$$

Outcomes within the same cluster are positively associated — they tend to move in the same direction relative to the overall mean.

1.5 Cluster-Specific Distribution

We can define $X_{ig} = \{X_i : \text{individual } i \in \text{cluster } g\}$. X_{ig} can be thought of as a different random variable than X_i , with its own distribution. It is like zooming in a cluster, the probability of a certain value occurring would be different than the probability of the same value occurring in the whole population. It has its own expectation:

$$\mathbb{E}(X_{ig}) = E(X_i \mid X_i \in g) \equiv \mu_g$$

And, due to the law of total expectation:

$$\mathbb{E}(\mu_g) = \mathbb{E}[\mathbb{E}(X_i \mid X_i \in g)] = E[X_i]$$

It also has its own variance, which is the conditional variance of X_i , given that $X_i \in g$. Call G the event $X_i \in g$

$$\begin{aligned}\sigma_g^2 &\equiv \mathbb{E}[X_{ig} - \mathbb{E}(X_{ig})]^2 \\ &= \mathbb{E}[(X_i - \mathbb{E}(X_i \mid G))^2 \mid G] \\ \text{Var}(X_i \mid G) &= \mathbb{E}[X_i^2 \mid G] - (\mathbb{E}[X_i \mid G])^2\end{aligned}$$

Due to the law of total variance:

$$\text{Var}(X_i) = \mathbb{E}[\text{Var}(X_i \mid G)] + \text{Var}(\mathbb{E}[X_i \mid G]).$$

2 Het

There are heterogeneous effect across clusters, this means that the true parameter β is different in every cluster.

$$y_{ig} = \alpha + \beta_g X_{ig} + u_{ig} \tag{1}$$

assume u_{ig} has a cluster component and an individual-and-cluster component:

$$u_{ig} = c_g + v_{ig}$$

Where $v_{ig} \sim (0, \sigma_v^2)$, $c_g \sim (0, \sigma_c^2)$ but of course c_g varies only at the cluster level.

In this way the expected value of the error given that individual i is in cluster g is:

$$\mathbb{E}_g(u_{ig}) = c_g$$

But as long as we have an intercept we can assume w.l.o.g that $\mathbb{E}[u_{ig}] = 0$. If I estimate $\hat{\beta}_{\text{ols}}$ inside cluster g , I get:

$$(\hat{\beta}_g - \beta_g) = \left(\sum_{i \in g} X_i^2 \right)^{-1} \left(\sum_{i \in g} X_i u_i \right)$$

And:

$$\hat{\beta}_g \stackrel{a}{\sim} \mathcal{N}\left(\beta_g, \frac{1}{N_g} \left(\sum_{i \in g} X_i^2\right)^{-1} B \left(\sum_{i \in g} X_i^2\right)^{-1}\right)$$

where B_g is:

$$B_g = \text{Var} (X_{ig} u_{ig})$$

If I estimate β across G clusters I get a vector of G $\hat{\beta}_g$, and for each of these I can consistently estimate an asymptotic variance using White variance estimator since observation inside a cluster are i.i.d. Since a linear combination of normal is itself normal I can average out those variances and get an estimate of the Avar of $\hat{\beta}_{\text{ols}}$.

To test this let us simulate a dataset, first we draw a random vector of betas from a normal distribution with mean 2 and variance 4.

```
include("../src/ClusterSimulation.jl")
G = 50 #Number of clusters
Ng = 1000 # Number of individuals per cluster

β = rand(Normal(2,4),G)
```

```
50-element Vector{Float64}:
 5.088698411760452
-2.2818348234475545
-2.2726481655182917
 2.1791495740744162
 2.017374523191644
-0.7340100200599364
 6.058188429783367
-1.130487982952011
 1.984427111499217
 4.129252462954926
 6.829300574831187
 2.8007510101635997
 3.073695368115863
 ⋮
 2.832775506289823
 2.948130605289181
 7.459529767510888
-5.747963710987246
 1.7729263920930463
 0.22795987595287648
 5.232265249603352
-1.1898560329598653
 0.7266460864625144
-0.03449602073321145
```

```
-0.8212976607627436  
4.639093868519023
```

Then we use each of these betas to simulate a cluster according to the model in Equation 1

```
pop = hePopulation(G,β,randn(G))
```

```
hePopulation(50, [5.088698411760452, -2.2818348234475545, -2.2726481655182917,  
2.1791495740744162, 2.017374523191644, -0.7340100200599364, 6.058188429783367,  
-1.130487982952011, 1.984427111499217, 4.129252462954926 ...  
7.459529767510888, -5.747963710987246, 1.7729263920930463,  
0.22795987595287648, 5.232265249603352, -1.1898560329598653,  
0.7266460864625144, -0.03449602073321145, -0.8212976607627436,  
4.639093868519023], [-0.9955677184370894, -0.1420458712559008,  
1.6215086524998816, 0.543625888869309, -0.944699407079914,  
-0.06048734406431664, 2.1237004993302606, -0.3500507221858935,  
0.8607655359232985, -0.29330750025228647 ... 0.09507807445457324,  
-0.4992009200348004, 1.94339751310204, -2.0828523077585244,  
1.5357652714448689, -0.8528815312624796, -1.2765538623131578,  
-1.6770139354283735, -0.8461985019295667, 0.6406935965381926])
```

This line creates a heterogeneous-effects population.

- **G** is the number of clusters.

- **β** is the vector of betas.

- **randn(G)** produces the vector $c_g \sim \mathcal{N}(0, 1)$.

The constructor returns a population struct that stores the cluster-level parameters used later to simulate data.

```
sam = sample(pop, Ng)
```

```
Cluster sample struct
```

This line generates a sample from the population.

- **Ng** is the number of observations per cluster.

The function draws data for all **G** clusters, each containing **Ng** observations, and returns a sample struct built according to the population's parameters.

The sample struct has a field **allclusters** that is an array of cluster structs, so we can define a function that compute $\hat{\beta}_g$ in each cluster and then broadcast it to all clusters in our sample, getting **G** estimates.

```
betahat = permutedims(stack(bols.(sam.allclusters))[:,2])
```

```
50-element Vector{Float64}:
```

```
 5.077188838846604
 -2.283725898697826
 -2.2693359546984784
 2.231740685015916
 2.0309065087551765
 -0.7241860615512835
 6.020300210394592
 -1.1153016650822691
 1.9536896704027733
 4.1315847666443615
 6.851399225652141
 2.8090047917094823
 3.0780340446501264
 ⋮
 2.846951482305467
 2.965025090034696
 7.4901422505775095
 -5.710957885647664
 1.793767434703426
 0.2592551544486365
 5.225033622352911
 -1.1681477743675772
 0.7496262659158976
 -0.0197367450146657
 -0.7824806468189728
 4.608267385711585
```

And similarly define a function that compute the White Variance estimator and broadcast it to all clusters.

```
WhiteAvar.(sam.allclusters)
```

```
50-element Vector{Matrix{Float64}}:
```

```
[0.000997914237318755 -6.080670391948711e-5; -6.080670391948707e-5
 0.001013413686982218]
[0.0010790837440304208 7.942834825158578e-5; 7.942834825158578e-5
 0.0011169281287722714]
[0.0009911660783528066 1.3998363075139554e-5; 1.3998363075139502e-5
 0.000955780688543717]
[0.0010204088867088853 3.733795557308652e-5; 3.7337955573086484e-5
 0.0010515932534936143]
[0.0009262661495312938 -9.814194906658057e-5; -9.814194906658064e-5
 0.0009012058752843286]
[0.0009910588513764253 -7.10083397826675e-5; -7.100833978266753e-5
 0.0009175465477991379]
[0.0009503913986314884 -1.756157499107687e-5; -1.7561574991076894e-5
```



```

0.0009393746355060233]
[0.0010067082594194456 1.0196221772817936e-5; 1.019622177281792e-5
0.0008418638509832287]
[0.0009725886532468074 -7.865069850094322e-5; -7.86506985009432e-5
0.0009817722501544591]
[0.0009960141994013211 5.307735249463491e-5; 5.307735249463488e-5
0.0010552714453648236]
[0.0010603392767801565 8.129201467067948e-7; 8.129201467067871e-7
0.0010353184272005103]
[0.000922660801862445 1.528130459470851e-5; 1.528130459470843e-5
0.000957781071933039]
[0.0010265665884436833 2.8872600655200553e-5; 2.887260065520049e-5
0.0010429506666186441]
:
[0.00101296355474145 -5.486751059428583e-6; -5.4867510594286065e-6
0.0009552818141518117]
[0.0010104886414024838 1.2400234435048258e-5; 1.2400234435048231e-5
0.0009927853387456996]
[0.0010163169387641944 -6.339626184633036e-5; -6.339626184633036e-5
0.0009877190728531104]
[0.0010430116258394002 3.7357008792990846e-5; 3.735700879299093e-5
0.00117482272477717]
[0.0010206367693159372 -2.7648406431903676e-5; -2.764840643190369e-5
0.0011021841498293546]
[0.0010268634522203734 -4.435533774011582e-5; -4.435533774011585e-5
0.0009851442511022441]
[0.0009645954629093965 6.0951151862596195e-5; 6.095115186259621e-5
0.0009178061635711255]
[0.0009286363346017035 8.142953427103056e-6; 8.142953427103063e-6
0.0008931983571125336]
[0.0010311787182974764 -3.246159870155027e-5; -3.246159870155023e-5
0.001111208451909863]
[0.0009899657457548214 1.3885742426842784e-5; 1.3885742426842826e-5
0.0010166857847301238]
[0.0009648706144488252 7.15312188936729e-7; 7.153121889368764e-7
0.0010375742597426832]
[0.0009713931039424013 8.38433120166976e-5; 8.384331201669756e-5
0.0010325193047199008]

```

We are interested in the average across clusters of β_g , the average partial effect. Note that since we have a finite amount of clusters, each of them with the same size, the APE would be:

$$\beta_{\text{APE}} \frac{1}{G} \sum_g \beta_g$$

Note that doing $\frac{1}{G} \sum_g \hat{\beta}_g$ is the same as computing $\hat{\beta}_{\text{ols}}$ in the whole sample. We define a function to do so:

```
function bols(s::clSample)
    X = getallX(s)
    Y = vcat(getallY(s)...)
    betahat = inv(X' * X) * (X'Y)
end
```

```
bols (generic function with 2 methods)
```

3 Consistency as $N_g \xrightarrow{p} \infty$

We want to test whether our estimate of the average partial effect gets better if the number of people that we draw in each cluster increases. In order to do so we ran a series of monte-carlo simulation increasing N_g .

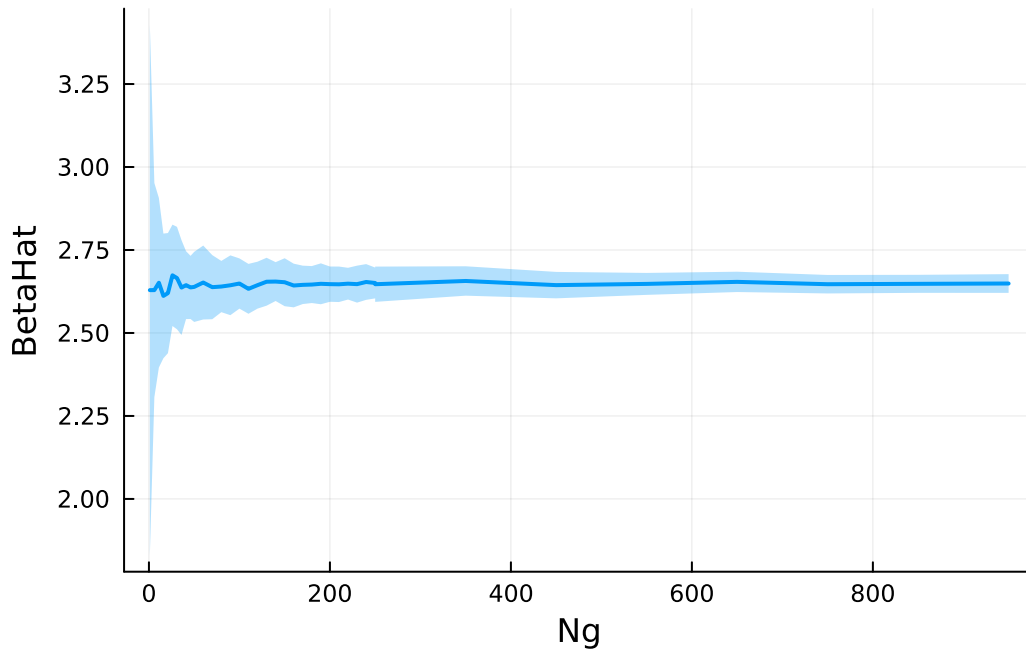


Figure 1: Consistency of betahat as N_g

What we see in Figure 1 is the result of the monte-carlo simulation, the line is the average estimates for each N_g , while the shaded area is the standard deviation, we can clearly see from the graph that as the number of observations inside a cluster grows $\hat{\beta}_{ols} \xrightarrow{p} \beta_{APE}$.

Bibliography

- [1] A. C. Cameron and D. L. Miller, “A Practitioner's Guide to Cluster-Robust Inference,” *Journal of Human Resources*, vol. 50, no. 2, pp. 317–372, Mar. 2015, doi: 10.3368/jhr.50.2.317.

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